



When Should IPL Teams Attack?

*Predicting the Aggression-Risk
Trade-off in IPL Chases*

Phase 2 – ORU IPL Analytics Hackathon

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The Free Lunch

2×

Sixes per match doubled
(9 → 18 over 18 seasons)

↓ 30%

Dot balls dropped
significantly

≈ 12

Dismissals stayed FLAT
for all 18 years

Teams got twice as aggressive but didn't get out more.

It looked like a free lunch.

Phase 2 asks: Is it really free lunch?

THE INVESTIGATION

Training a Model to Watch Cricket

130,282

Balls analyzed
(Innings 2 only)

1,088

Chase matches

10

Engineered features

Runs scored & needed — Progress + distance to target

Wickets in hand — Resources remaining

Balls remaining — Time pressure

rr_gap — **PRESSURE: how much faster must you score?**

Phase (1/2/3) — Powerplay / Middle / Death

Momentum — Runs in last 6 balls — rhythm

MODEL SELECTION

300 Cricket Experts Voting

Random Forest = 300 trees, each analyzing the match differently, then majority vote.

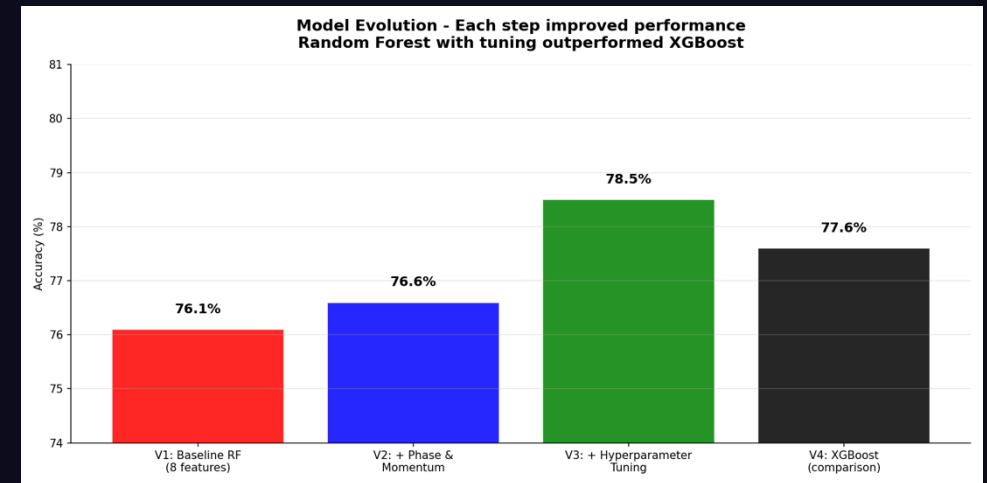
V1: Baseline RF | 8 features, defaults | 76.1%

V2: + Features | Added phase & momentum | 76.6%

V3: + Tuning ✓ | 108 combos × 3-fold CV | 78.5%

V4: XGBoost | Gradient boosting | 77.6%

Train: 2007/8–2022 → Test: 2023–2025



WHAT DRIVES WINNING?

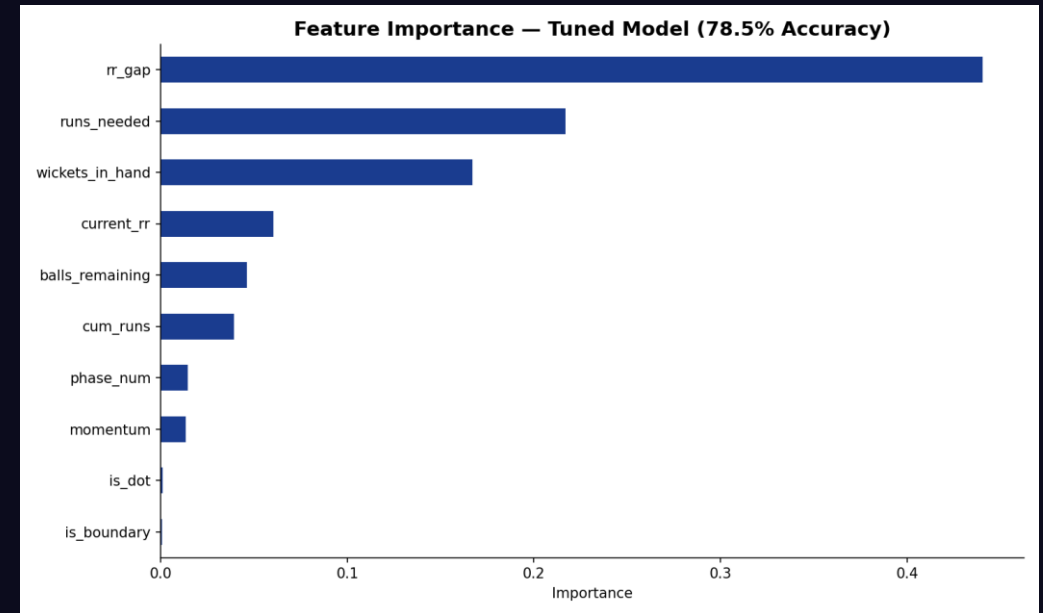
Pressure Is Everything

44% of prediction importance comes from rr_gap (pressure)

Not individual shots. Not the over number.

The cumulative pressure state.

is_boundary and is_dot have near-zero importance.
One six doesn't win a match. A growing pressure gap loses one.



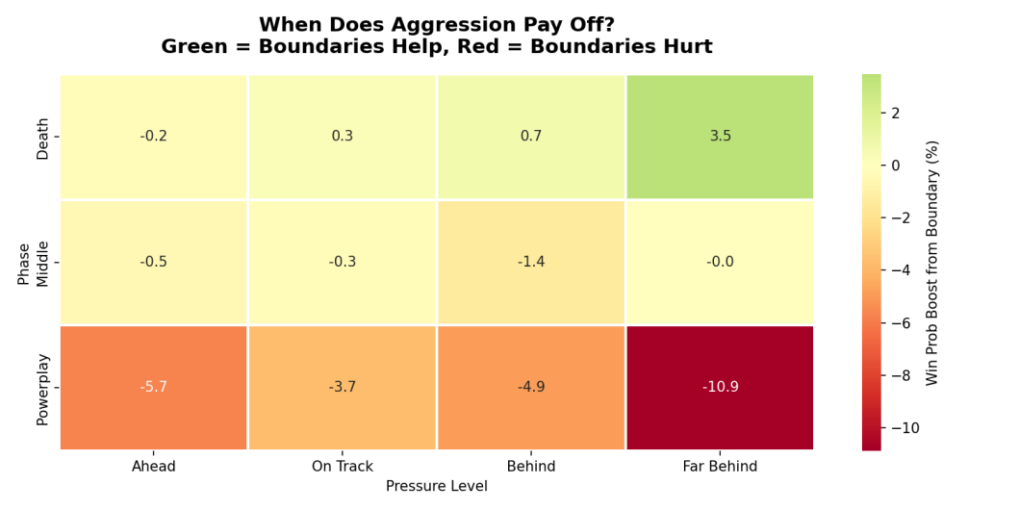
THE PLOT TWIST

Aggression is NOT Equal Everywhere

Phase	Pressure	Boundary Effect	Strategy
Death overs	Far behind	+3.5%	ATTACK
Death overs	On track	+0.3%	Neutral
Middle overs	On track	-0.3%	Neutral
Powerplay	Ahead	-5.7%	Defend
Powerplay	Far behind	-10.9%	DEFEND

Hitting a boundary in the powerplay makes you MORE likely to LOSE.

Because you're risking wickets — and early wickets are catastrophic.



CAN YOU TRUST IT?

The Model Is Honest

81.1%

Precision — when it says "attack,"
it's right 81 out of 100 times

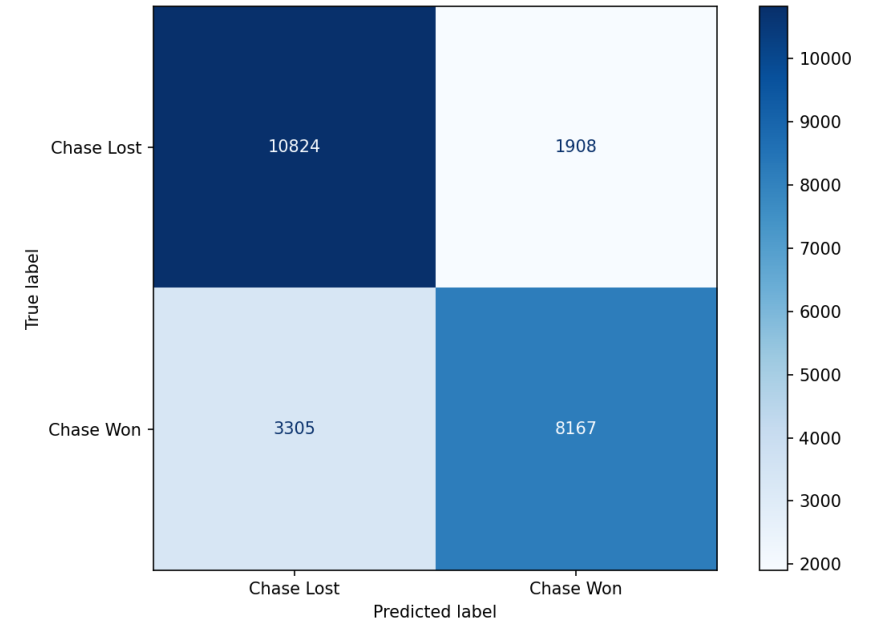
1.73×

Makes 1.73× more SAFE mistakes
than dangerous ones

99% → 69%

99% accurate in clear situations
69% in genuine toss-ups — honest

Confusion Matrix - Tuned Random Forest
Prediction matters more than recall here



SEE IT IN ACTION

Live Demo



Comfortable Chase

120/2 chasing 150, over 16
→ 90.6% win probability



Powerplay Disaster

15/4 chasing 175, over 4
→ 0.5% win probability

Includes "What If" analysis: boundary vs dot ball vs wicket impact

18 years - 130,000 balls - 1 model - 1 strategy

Survive early.

Attack late.

*The IPL aggression revolution was real.
But the lunch is only free in the death overs.
In the powerplay, it costs you -10.9% per boundary.*



PBKS! WIN!

KINGS OF PUNJAB

CHAMPIONS

KINGS OF PUNJAB

RESULT: PBKS WON BY 16 RUNS

CHAMPIONS

PUNJAB KINGS
WIN FINAL MATCH!

PBKS: 188/6 (20) | OPPS: 172/9

| CHAMPIONS! | CHAMPIONS!

IPL 2024 FINAL | MOHALI

IPL 2024 FINAL | MOHALI

IPL 2024 FINAL | MOHALI

13
PRABHSIMRAN
SINGH

09
LIVINGSTONE

1
RABADA

AIWIGISHA



THANK YOU

*Special Thanks to the
entire ORU Community*